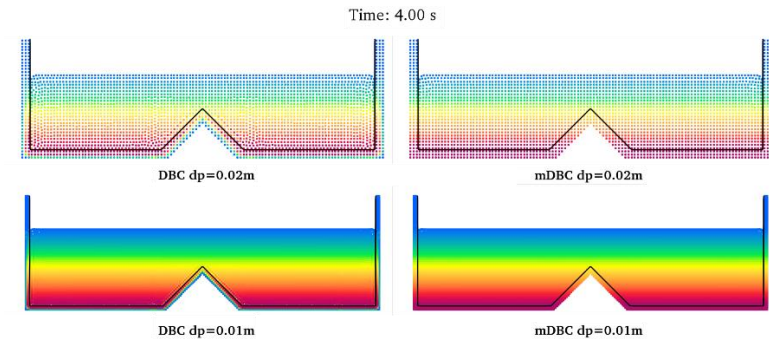


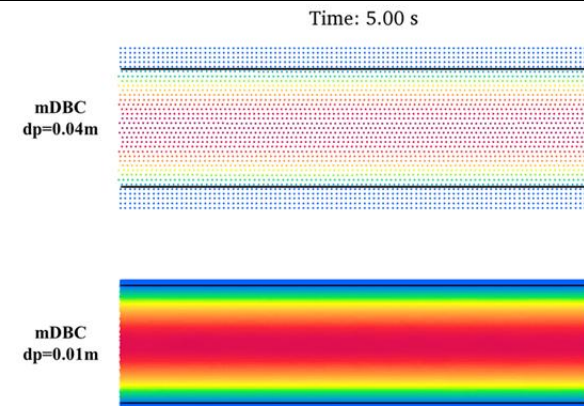
### 01\_STILLWEGE

- A 2-D still water tank encloses a trigonal wedge in the bottom centre of the tank with. A low value of artificial viscosity is used to observe better the effects and noise generated by the different boundary conditions (DBC and mDBC).
- Includes execution with DBC, mDBC SlipMode1: Vel0, and mDBC SlipMode2: No-Slip  
[Video](#)



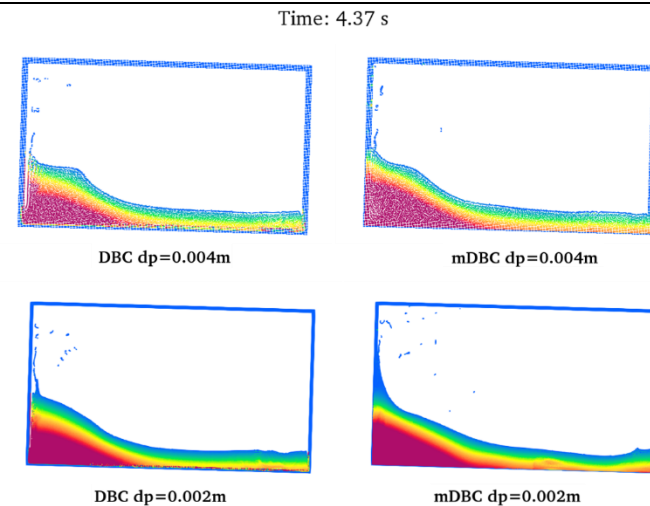
### 2\_POISEUILLE

- Poiseuille flow is simulated with the boundary walls located at  $z=\pm 0.5\text{m}$ , the maximum velocity is  $U=1\text{m/s}$  and the viscosity is  $\nu=0.1\text{m}^2/\text{s}$  resulting in a Reynolds number of  $Re=10$ .
- Includes execution with DBC, mDBC SlipMode1: Vel0, mDBC SlipMode2: No-Slip, and mDBC SlipMode3: FreeSlip  
[Video](#)
- Run Down flow is with boundary walls at  $z=\pm 0.5\text{m}$  and initial velocity  $U=1\text{m/s}$  with no body forces to drive the flow, the flow is then slowed down by the chosen slip condition. Includes execution with mDBC SlipMode2: No-Slip, and mDBC SlipMode3: FreeSlip



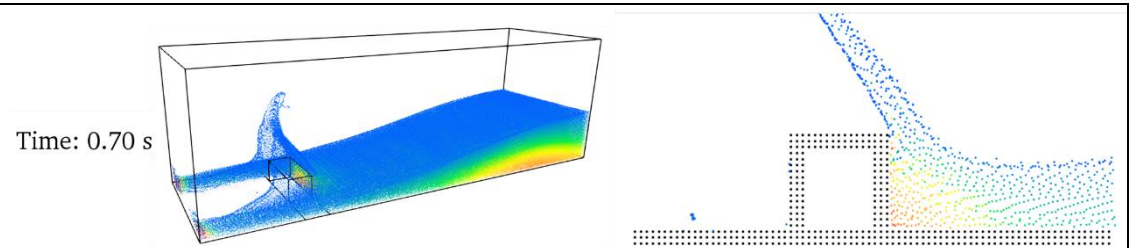
### 3\_SLOSHING

- The case will reproduce a sloshing experiment that includes moving boundaries: SPHERIC Benchmark Test Case #10
- Includes execution with DBC, mDBC SlipMode1: Vel0, mDBC SlipMode2: No-Slip and SlipMode2: No-Slip with No-Penetration activated  
[Video](#)



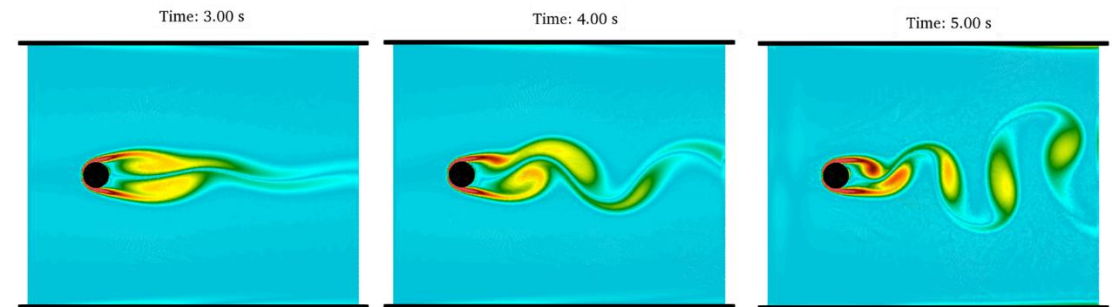
#### 04\_DAMBREAK

- 3-D dam break experiment that includes moving boundaries proposed as SPHERIC Benchmark Test Case #2.
- Includes execution with mDBC SlipMode1: Vel0 and mDBC SlipMode2: No-Slip with no-Penetration activated  
[Video](#)



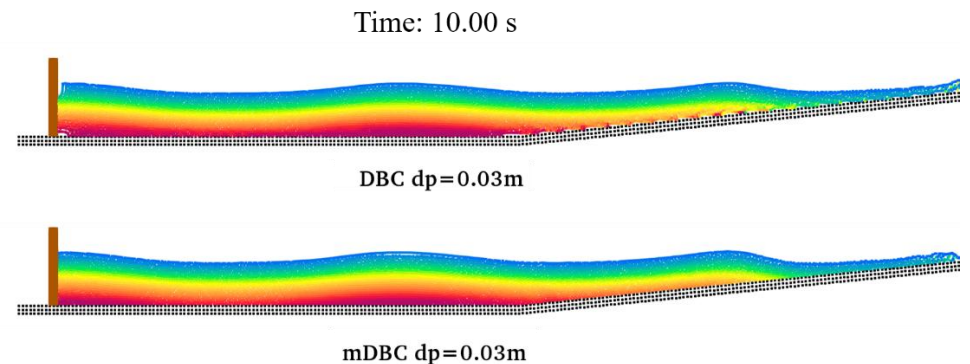
#### 05\_FLOWCYLINDER

- 2-D flow passing a cylinder of diameter  $D=0.2\text{m}$ , which is surrounded by a viscous fluid. Dimensions of the fluid domain are chosen to minimise boundary effects. The fluid is initialized with a constant velocity of  $U=1\text{m/s}$  and with  $Re=200$ .
- The circle is created using the Free Draw Mode instead of the Cartesian grid.
- Includes execution with DBC, mDBC SlipMode1: Vel0, mDBC SlipMode2: No-Slip  
[Video](#)



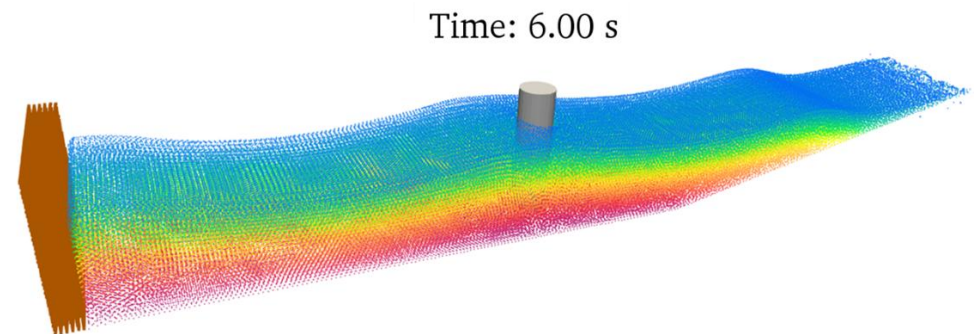
#### 06\_WAVETANK

- 2-D regular waves are generated and propagated in a numerical wave flume using DBC and the new mDBC.
- The beach is created using the Free Draw Mode instead of the Cartesian grid.
- Includes execution with DBC, mDBC SlipMode1: Vel0, mDBC SlipMode2: No-Slip and SlipMode2: No-Slip with No-Penetration activated  
[Video](#)



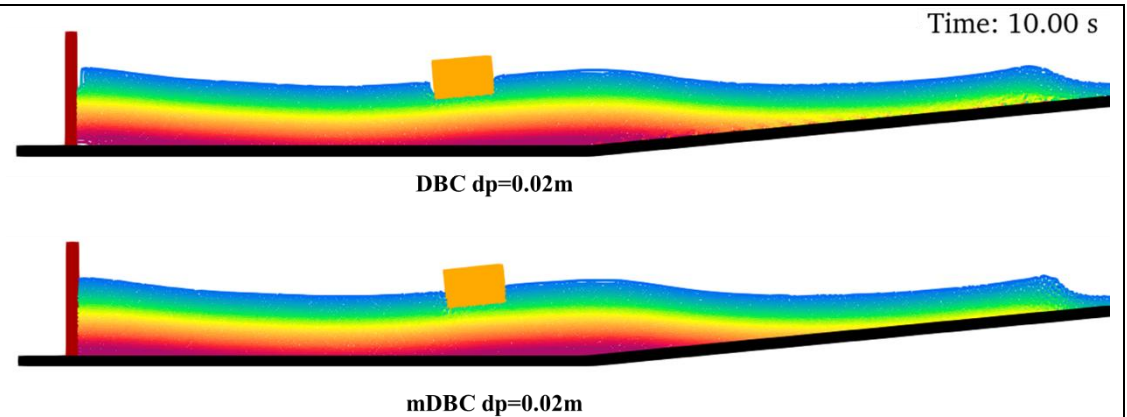
#### 07\_WAVESCYLINDER

- 3-D regular waves ( $H=0.1\text{m}$ ,  $T=1.2\text{s}$ ,  $d=0.5\text{m}$ ) passing a cylinder of diameter  $D=0.2\text{m}$  and  $0.7\text{m}$  high located in the middle of a wave flume.
- The cylinder is created using the Free Draw Mode instead of the Cartesian grid
- Includes execution with mDBC SlipMode1: Vel0, mDBC SlipMode2: No-Slip  
[Video](#)



## 08\_FLOATINGWAVES

- 2-D floating box under the action of regular waves with piston.
- The beach is created using the Free Draw Mode instead of the Cartesian grid.
- mDBC is applied not only to the tank and piston, but also to the floating box.
- Includes execution with DBC, mDBC SlipMode1: Vel0, mDBC SlipMode2: No-Slip [Video](#)
- 2-D floating box under the action of regular waves with piston with validation data (motions of the box) from [Ren et al., 2015].
- Includes execution with mDBC SlipMode1: Vel0 and mDBC SlipMode2: No-Slip



## 09\_FLOATINGDUCK

- 3-D floating external STL (duck) in a square tank using DBC and mDBC.
- mDBC is applied to an external STL using advanced tools during pre-processing in order to create normal vectors and to fill the geometry with floating particles.
- Includes execution with DBC, mDBC SlipMode1: Vel0 applied to duck only, and SlipMode2: No-Slip with No-Penetration activated applied to all boundaries [Video](#)

